# Portfolio Optimization & Factor Analysis

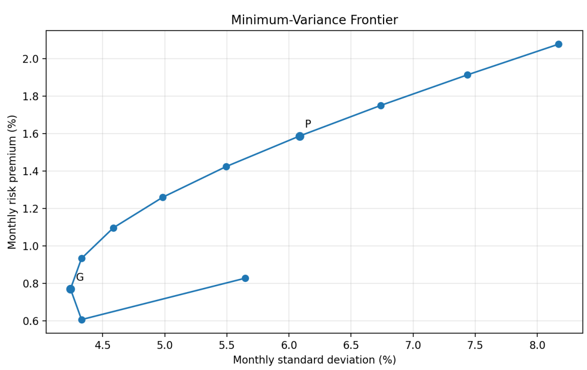
Mean-variance optimization, benchmark comparison, and factor-model analysis

## Objective

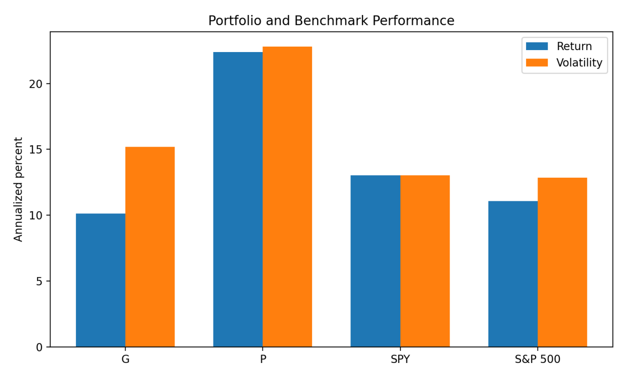
Construct global minimum-variance and maximum-Sharpe portfolios from eight equities, map a ten-point efficient frontier, compare performance with SPY and the S&P 500, and evaluate the optimized portfolio using common asset-pricing models.

## Headline findings

- Maximum-Sharpe portfolio: 22.4% annualized return, 22.8% volatility, and 0.925 in-sample Sharpe ratio.
- SPY: 13.0% annualized return, 13.0% volatility, and 0.900 Sharpe ratio.
- The optimized portfolio's Sharpe advantage was only about 0.025 and required short and concentrated positions.
- Fama-French and Carhart models explained only about 2%-3% of monthly optimized-portfolio return variation.
- The market regression was corrected so portfolio excess return is the dependent variable.




**Efficient frontier and optimized portfolios**




**Annualized return and volatility comparison**

## Corrections and limitations

The portfolio version corrects a mislabeled standard-deviation row, clarifies excess kurtosis and Capital Allocation Line terminology, reverses the original market regression to the proper asset-pricing orientation, and avoids claiming that positive in-sample alpha proves skill or underpricing. The analysis remains in-sample and excludes transaction costs, short-borrow constraints, taxes, and out-of-sample validation.